

LAB 6

Task # 01

```
import java.io.File;
import java.io.FileNotFoundException;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        // Specify the file name or path
        String fileName = "example.txt";

        // Call the method to read the file
        readFile(fileName);
    }
    public static void readFile(String fileName) {
        // Create a File object
        File file = new File(fileName);

        try {
            // Try to read the file using a Scanner object
            Scanner fileScanner = new Scanner(file);

            // Print the contents of the file (just an example)
            while (fileScanner.hasNextLine()) {
                System.out.println(fileScanner.nextLine());
            }

            fileScanner.close();
        } catch (FileNotFoundException e) {
            // Handle the exception and print an error message
            System.out.println("Error: The file " + fileName + " was not found.");
        }
    }
}
```

Task # 02

```
public class Main {
    public static void main(String[] args) {
        // Example integers
        int numerator = 10;
        int denominator = 0; // This will cause division by zero

        // Call the method to divide
```

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```
        divide(numerator, denominator);
    }

    public static void divide(int numerator, int denominator) {
        try {
            // Perform division
            int result = numerator / denominator;
            System.out.println("Result: " + result);
        } catch (ArithmeticException e) {
            // Handle division by zero (ArithmeticException)
            System.out.println("Error: Cannot divide by zero.");
        }
    }
}
```

Task # 03

```
public class Main {

    public static void main(String[] args) {
        // Attempt to create an array with a negative size
        int size = -5; // Negative size to trigger the exception

        // Call the method to create the array
        createArray(size);
    }

    public static void createArray(int size) {
        try {
            // Attempt to create an array with a negative size
            int[] array = new int[size];
            System.out.println("Array created with size: " + size);
        } catch (NegativeArraySizeException e) {
            // Handle the NegativeArraySizeException
            System.out.println("Error: Cannot create an array with a negative size.");
        }
    }
}
```

Task # 04

```
public class Main {

    public static void main(String[] args) {
        // Attempt to create an array with a negative size
        int size = -5; // Negative size to trigger the exception
```

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```
// Call the method to create the array
createArray(size);
}

public static void createArray(int size) {
    try {
        // Attempt to create an array with a negative size
        int[] array = new int[size];
        System.out.println("Array created with size: " + size);
    } catch (NegativeArraySizeException e) {
        // Handle the NegativeArraySizeException
        System.out.println("Error: Cannot create an array with a negative size.");
    }
}
}
```

Task # 05

```
// Define the custom checked exception

class InvalidAgeException extends Exception {

    // Constructor to initialize the exception with a message

    public InvalidAgeException(String message) {

        super(message);

    }

}
```

```
public class AgeValidator {

    public static void main(String[] args) {

        // Test with an invalid age

        int age = -5; // Invalid age for demonstration

        try {

            // Call method to validate age
```

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```
        validateAge(age);
    } catch (InvalidAgeException e) {
        // Catch and handle the custom exception
        System.out.println("Error: " + e.getMessage());
    }

    // Test with a valid age
    age = 25;

    try {
        // Call method to validate age
        validateAge(age);
        System.out.println("Age is valid: " + age);
    } catch (InvalidAgeException e) {
        // Catch and handle the custom exception
        System.out.println("Error: " + e.getMessage());
    }
}

// Method to validate age
public static void validateAge(int age) throws InvalidAgeException {
    // If the age is less than 0 or greater than 120, throw an exception
    if (age < 0 || age > 120) {
        throw new InvalidAgeException("Age must be between 0 and 120.");
    }
}
}
```